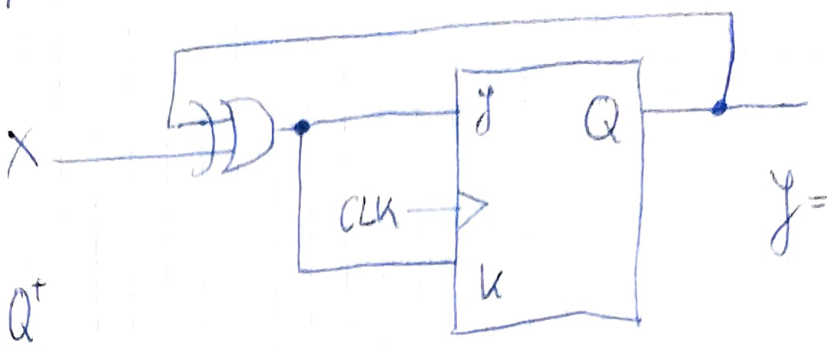
zadanie 1



$J = K = Q \oplus x$

XQ	JkQ+
00	000
01	110
10	111
11	001

X	0	1	0	0	1	1	0	1	0	0	1
Q:	0	0	1	0	0	1	1	0	1	0	0
Stan początkowy 0 Q:	1	0	1	0	0	1	1	0	1	0	0

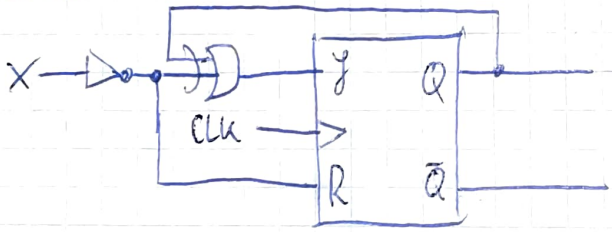
Stan początkowy 1

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zadanie 2.

$R = \bar{x}$

$S = Q \oplus \bar{x}$



1) CLK



x



Q



2) CLK



x

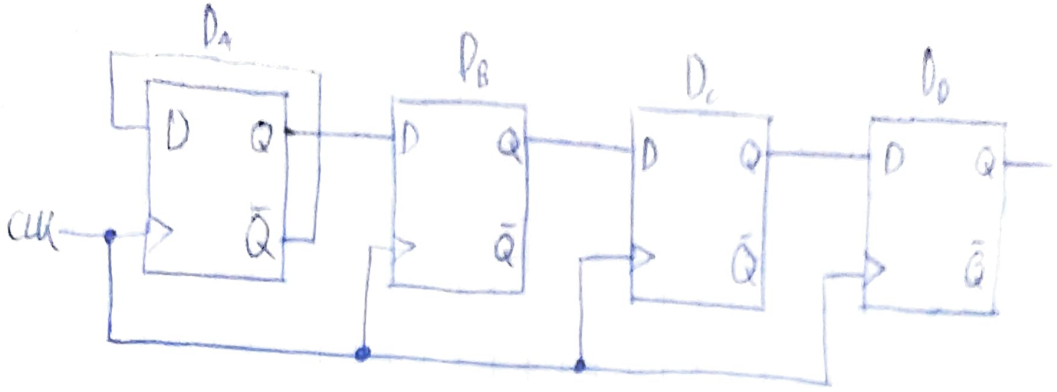


Q



▒▒▒ - Stany zakazane

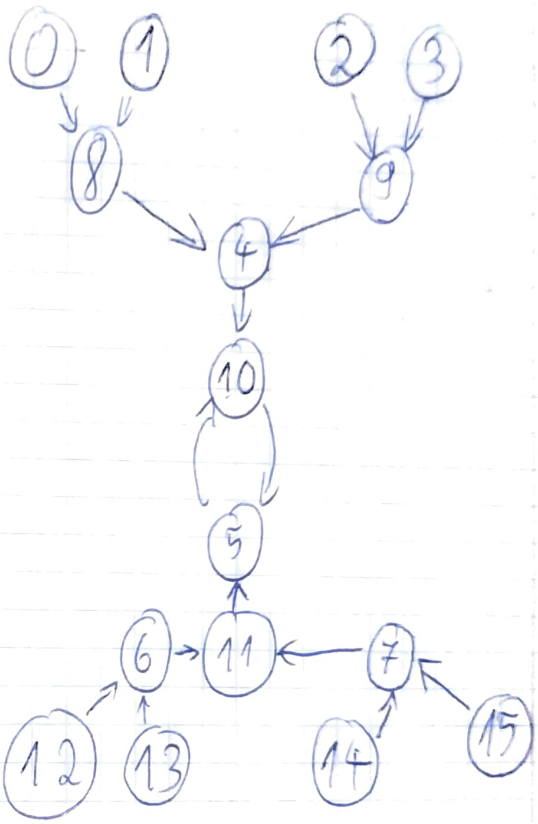
zadanie 3.



Q_A	Q_B	Q_C	Q_D	Q_A^+	Q_B^+	Q_C^+	Q_D^+
0	0	0	0	1	0	0	0
0	0	0	1	1	0	0	0
0	0	1	0	1	0	0	1
0	0	1	1	1	0	0	1
0	1	0	0	1	0	1	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	1	1
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	0	1	0	1	0	0
1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0
1	1	0	0	0	1	1	0
1	1	0	1	0	1	1	0
1	1	1	0	0	1	1	1
1	1	1	1	0	1	1	1

$D_A = \bar{Q}_A$

graf:



maksymalna liczba taktów zegara wymagana do stanu zawartości licznika to 3

Licznik jest bezpieczny.